

WARPSimLab

Release Notes

Version 4.3

Release Date: September 5, 2026

Overview

Version 4.3 improves input validation, portfolio data entry, simulation performance, and internal architecture, and adds expanded HSA functionality.

Changes and Improvements

- Completed validation of user inputs throughout the GUI.
 - Added additional validation at the GUI/simulation-core interface.
 - Expanded HSA handling for income and expenses.
 - Added percentage-based portfolio input and updated all portfolio input dialogs.
 - Added support for annual fixed-percentage increases to Special Income.
 - Improved performance of comparison reports that use Historical Windows. Selected reports now use sampled historical windows to reduce processing time while retaining full-resolution analysis where appropriate.
 - Refactored major portions of the GUI and simulation core to improve maintainability.
 - Split `gui_init.py` into smaller functional components.
 - Split the simulation core into separate income/expense and income/withdrawal processing paths.
 - Split `gui_simulationController.py` into smaller functional components. This prepares for future Scenario Explorer development.
 - Expanded core testing and performed additional simulation cleanup and fine-tuning.
 - Fixed numerous minor bugs.
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Known Limitations

- Results are generated from simplified mathematical models. For example, the current model assumes that 100% of Social Security benefits are taxable. After-tax equity realized gains and dividends are combined and treated as qualified long-term capital gains.

- Simulations simplify financial markets and individual financial situations and do not reproduce every real-world condition.
 - Output should not be interpreted as predictions or recommendations.
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Future Development

- Add explicit future health expense options.
- Add additional retirement withdrawal options.
- Redesign and reimplement Scenario Explorer.
- Add additional pension growth and adjustment options.
- Add scalable GUI fonts to improve application scaling across different display sizes and resolutions.
- Expand documentation.